Breadth First Search (BFS)

```
#include <stdio.h>
#define MAX 50

int main()
{
    int graph[MAX][MAX];
    int visited[MAX];
    int queue[MAX];
    int n, i, j;
    int front = 0, rear = 0;
    int start, current;

    // Read number of vertices
    printf("Enter number of vertices: ");
    scanf("%d", &n);

    // Read adjacency matrix
    printf("Enter the adjacency matrix:\n");
    for (i = 0; i < n; i++)
    {
        for (j = 0; j < n; j++)
        {
            scanf("%d", &graph[i][j]);
        }
    }

    // Initialize visited array
    for (i = 0; i < n; i++)
    {
        visited[i] = 0;
    }

    // Read starting vertex
    printf("Enter starting vertex: ");
    scanf("%d", &start);

    // Insert starting vertex into queue
    visited[start] = 1;
    queue[rear] = start;
    rear++;

    printf("BFS Traversal: ");
```

```
    // BFS traversal
    while (front < rear)
    {
        // Remove vertex from queue
        current = queue[front];
        front++;
        printf("%d ", current);

        // Find adjacent vertices
        for (i = 0; i < n; i++)
        {
            if (graph[current][i] == 1 && visited[i] == 0)
            {
                visited[i] = 1;
                queue[rear] = i;
                rear++;
            }
        }
    }

    return 0;
}
```

Output:

```
Enter number of vertices: 5
Enter the adjacency matrix:
0 1 1 1 0
1 0 1 0 0
1 1 0 0 1
1 0 0 0 0
0 0 1 0 0
Enter starting vertex: 0
BFS Traversal: 0 1 2 3 4
```